

Software Requirements Specification

Event Management System

Version: 1.0

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Project: A system to manage and organize events with registrations.

(EVENT MANAGEMENT SYSTEM)

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Revision History

Version	Date	Author	Change Summary
1.0	12-09-2026	Project Team	Initial Event Management System SRS prepared using the supplied SRS template structure.

Approvals

Role	Name	Signature / Email	Date
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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification for the Event Management System. It defines the functional and non-functional requirements, interfaces, quality expectations, security requirements, acceptance criteria and traceability needed to guide implementation and evaluation of the project.

1.2 Scope

The system is a web-based platform for managing events and participant registrations. It supports event creation and publication by organizers, event discovery by participants, registration and cancellation, organizer visibility into registrations, and administrative management.

The Version 1.0 scope intentionally excludes payment processing, specialized QR/barcode hardware, external calendar synchronization and production email/SMS infrastructure. These can be considered only through an explicit requirements change.

1.3 Audience

Developers, testers, project members, course instructors/evaluators and future maintainers.

1.4 Definitions

Term	Definition
SRS	Software Requirements Specification.
FR	Functional Requirement.
NFR	Non-Functional Requirement.
RTM	Requirements Traceability Matrix.
Participant	User who discovers events and registers for them.
Organizer	Users authorized to create and manage events.
Administrator	Privileged user who manages users/events and administrative information.
Registration	Participant's association with an event after successful registration.
Capacity	Maximum number of participants permitted for an event.

2. Overall Description

2.1 Product Perspective

The Event Management System is a new, self-contained web application. The proposed architecture uses a React.js frontend, a Node.js/Express.js backend and MongoDB for persistent application data. The frontend communicates with backend services through REST APIs; the backend applies business rules and accesses the database.

2.2 Major Product Functions (Detailed)

- User registration and authentication.
- Role-based authorization.
- Event creation, editing, publishing and cancellation.
- Event browsing, search and filtering.
- Event detail viewing.
- Participant registration and cancellation.
- Capacity and registration-deadline enforcement.
- Organizer registration count and participant-list viewing.
- Administrator user and event management.
- Application confirmations/notifications.
- Audit records for important administrative and security-related actions.

2.3 User Roles and Characteristics (Expanded)

Role	Characteristics / Goals
Participant	General web user; expects simple event discovery, clear event details and a short registration flow.
Organizer	Creates and manages events; needs accurate event status and participant information.
Administrator	Privileged system user; needs controlled access to user/event management and audit information.

2.4 Operating Environment

The system will run as a web application using React.js, Node.js/Express.js and MongoDB. The academic version will be demonstrated locally on team computers using a modern browser such as Chrome. No specialized hardware is required.

2.5 Constraints

- React.js shall be used for the frontend.
- Node.js/Express.js shall be used for the backend.
- MongoDB shall be used for persistent application data.
- The project shall remain within realistic mini-project scope.
- Server-side validation and authorization shall be used for protected operations.
- A shared version-control repository shall be used during development.

2.6 Assumptions and Dependencies

- Users have a supported browser and network connection.
- Backend and database services are available during normal operation.
- The system clock is correctly configured for event dates and registration deadlines.
- Version 1.0 does not depend on an external email/SMS provider.
- The project demonstration is performed under controlled local/academic load.

3. External Interface Requirements

3.1 User Interfaces

The primary interfaces shall include registration/login, event listing, search/filter controls, event details, organizer event-management screens, participant registration history and administrator management screens. Forms shall provide clear validation and error feedback.

3.2 Hardware Interfaces

The Event Management System does not require dedicated or specialized hardware interfaces. It shall operate through standard laptops, desktops, tablets or smartphones with a supported web browser. No physical event-management device is controlled by Version 1.0.

3.3 Software Interfaces

The React.js frontend shall communicate with the Node.js/Express.js backend using REST APIs. The backend shall communicate with MongoDB through the database access layer. API data shall use JSON.

3.4 Communications

Client-server communication shall use HTTP-based REST APIs. HTTPS shall be used for deployed environments handling real credentials. The backend shall return appropriate responses for validation, authentication, authorization, conflict and server-error conditions.

4. System Features (Detailed)

4.1 User Registration & Authentication

Description and Priority: Users can create accounts and authenticate before accessing protected functions. Priority: High.

Stimulus/Response: A user submits registration or login information; the system validates it and creates/authenticates the account, then grants access according to role.

Functional Requirements:

- EMS-F-001: The system shall allow a new user to create an account using required registration details.
- EMS-F-002: The system shall authenticate registered users using valid credentials.
- EMS-F-003: The system shall reject invalid credentials and duplicate account identifiers with a clear error message.
- EMS-F-004: The system shall restrict protected functions to authenticated users.

4.2 Event Creation, Editing & Publishing

Description and Priority: Organizers can create and manage events with the information participants need. Priority: High.

Stimulus/Response: An authorized organizer enters event details; the system validates the information and saves the event in the appropriate lifecycle state.

Functional Requirements:

- EMS-F-005: The system shall allow an authorized organizer to create an event with required event details.
- EMS-F-006: The system shall allow an organizer to edit an event before it begins, subject to defined rules.
- EMS-F-007: The system shall allow an organizer to publish an event after required information is complete.
- EMS-F-008: The system shall allow an organizer to cancel an event and prevent new registrations for the cancelled event.

4.3 Event Discovery & Search

Description and Priority: Participants can browse published events and locate relevant events through search and filtering. Priority: High.

Stimulus/Response: A participant opens the event listing or enters search/filter criteria; the system returns published events matching the criteria.

Functional Requirements:

- EMS-F-009: The system shall display published events to participants.
- EMS-F-010: The system shall allow participants to search events using supported criteria such as title or category.
- EMS-F-011: The system shall allow participants to filter events using supported criteria such as date.
- EMS-F-012: The system shall display event details including title, description, schedule, venue/online information, capacity and registration deadline.

4.4 Event Registration & Cancellation

Description and Priority: Participants can register for available events and cancel registrations when permitted. Priority: High.

Stimulus/Response: A participant selects an event; the system checks authentication, deadline, capacity and duplicate registration before creating the registration.

Functional Requirements:

- EMS-F-013: The system shall allow an authenticated participant to register for an event when registration is open and capacity is available.
- EMS-F-014: The system shall prevent duplicate registration by the same participant for the same event.
- EMS-F-015: The system shall prevent registration after the deadline or when the event reaches capacity.
- EMS-F-016: The system shall allow a participant to cancel an existing registration when cancellation is permitted.
- EMS-F-017: The system shall update registration status and available capacity after valid cancellation.

4.5 Organizer Registration Management

Description and Priority: Organizers can monitor participation in events they own. Priority: High.

Stimulus/Response: An organizer opens an owned event; the system displays registration count and the authorized participant list.

Functional Requirements:

- EMS-F-018: The system shall allow an organizer to view the registration count for an owned event.
- EMS-F-019: The system shall allow an organizer to view the participant list for an owned event.
- EMS-F-020: The system shall prevent an organizer from accessing registration information for events they do not own unless authorized.

4.6 Administration & User/Event Management

Description and Priority: Administrators can perform system-level management activities. Priority: Medium.

Stimulus/Response: An administrator performs a permitted management action; the system verifies the Administrator role and applies the action.

Functional Requirements:

- EMS-F-021: The system shall allow an administrator to manage user accounts.
- EMS-F-022: The system shall allow an administrator to review and manage events when intervention is required.
- EMS-F-023: The system shall restrict administrator operations to the Administrator role.

4.7 Notifications & Audit Records

Description and Priority: The system provides useful confirmations and records important administrative/security actions. Priority: Medium.

Stimulus/Response: A relevant event occurs; the system provides status/notification feedback and records applicable audit information.

Functional Requirements:

- EMS-F-024: The system shall provide confirmation feedback after successful registration and permitted cancellation.
- EMS-F-025: The system shall notify affected users when an event is cancelled.
- EMS-F-026: The system shall record important administrative and security-sensitive actions with timestamps.
- EMS-F-027: The system shall not include passwords or authentication secrets in audit records.

5. Non-Functional Requirements (Detailed)

5.1 Performance Requirements

Req ID	Requirement	Category	Priority	Acceptance / Measurement
EMS-NF-001	Once the backend service is active (not in a cold-start state), event listing and normal registration actions should complete within about 3 seconds for most sampled requests, under normal demonstration load.	Performance	High	95% of sampled requests meet the target. Test: TC-PERF-01 Majority of sampled warm requests meet the ~3-second target. Test: TC-PERF-01
EMS-NF-002	Once the backend service is active, event search/filter operations should return results within about 2 seconds under normal demonstration load.	Performance	Medium	Sampled warm response times meet the ~2-second target. Test: TC-PERF-02
EMS-NF-003	Once the backend service is active, Organizer/admin dashboards should load within about 5 seconds under normal demonstration load.	Performance	Medium	Sampled warm dashboard load times meet the ~5-second target. Test: TC-PERF-03
EMS-NF-004	The system shall maintain registration data consistency so that capacity is not exceeded and duplicate active registrations are not created.	Reliability/Data Integrity	High	Edge/concurrency tests pass. Test: TC-DATA-01
EMS-NF-005	The system shall support the core participant and organizer workflows on representative desktop and mobile browser viewports.	Usability	Medium	Core workflows pass viewport tests. Test: TC-UX-01
EMS-NF-006	The system shall log relevant application/security events with timestamps sufficient for troubleshooting and audit review.	Maintainability/Audit	Medium	Required records can be retrieved. Test: TC-OPS-01
EMS-NF-007	Due to free-tier hosting limits, the first request after a period of backend inactivity may take up to 60 seconds while the service restarts; this cold-start delay is excluded from the performance targets above.	Performance	Low	Documented as a known limitation, not subject to pass/fail testing. Test: N/A

5.2 Safety Requirements

This academic web application does not directly control physical equipment or safety-critical hardware. Therefore, Version 1.0 has no specific physical safety hazards. As a basic safeguard against loss of application information, persistent data shall be stored in the database rather than only in temporary application memory.

5.3 Security

5.3.1 Security Objectives

- **SO-01** — Confidentiality: The system shall protect user credentials and registration information from unauthorized disclosure.
- **SO-02** — Integrity: The system shall prevent unauthorized or invalid modification of user, event and registration data.
- **SO-03** — Controlled Access: The system shall ensure that users can perform only operations permitted by their assigned role.

5.3.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance Criteria / Test
EMS-SR-001	The system shall store passwords only as salted one-way hashes and shall never store plaintext passwords.	Security	High	Database inspection shows no plaintext passwords. TC-SEC-01
EMS-SR-002	The system shall enforce server-side authorization checks for protected Organizer and Administrator operations.	Security	High	Unauthorized API calls are rejected. TC-AUTHZ-01
EMS-SR-003	The system shall validate user-controlled input on the server before processing or persistence.	Security	High	Malformed/injection-style inputs are safely rejected. TC-SEC-02
EMS-SR-004	The system shall not expose passwords, authentication tokens, secrets or internal stack traces in normal responses or logs.	Security	High	Response/log inspection passes. TC-SEC-03
EMS-SR-005	The system shall record security-relevant events such as repeated authentication failures and privilege changes with timestamps.	Security	Medium	Audit records contain required events. TC-AUDIT-01

EMS-SR-006	Administrator-only functions shall be inaccessible to Participant and Organizer roles.	Authorization	High	Role-boundary tests pass. TC-AUTHZ-02
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5.4 Software Quality Attributes

- Usability: the participant should be able to discover an event and register with minimal navigation; organizer/admin screens should be clear.
- Reliability: registration data shall remain consistent with no duplicate active registrations or over-capacity registrations.
- Maintainability: frontend, backend and database responsibilities shall remain modular and the team shall use a shared repository.
- Scalability: the design should permit additional users and events later without redesigning the core workflow.
- Security: authentication, authorization, input validation and safe credential handling shall be considered throughout implementation.

5.5 Business Rules

- Only authenticated users can register for events.
- Only authorized organizers can create and manage their own events.
- Only administrators can perform administrator-level management.
- A participant can have at most one active registration for a given event.
- An event cannot accept registrations after its registration deadline or after reaching capacity.
- A participant's registration history is visible only to that participant and authorized administrators.

6. Quality Attributes & Acceptance Tests

Acceptance is based on verification of requirements rather than visual completion alone.

6.1 Quality Targets

Quality Attribute	Target / Evidence
Performance	Defined response-time targets in Section 5.1 are met under normal demonstration load.
Reliability	Registration rules prevent duplicates and over-capacity conditions.
Security	Authentication, authorization, credential protection and safe error handling tests pass.

Usability	Core participant/organizer workflows remain usable on representative desktop/mobile viewports.
Maintainability	Major responsibilities are modular and traceable to implementation modules.
Data Integrity	Successful operations leave consistent event, user and registration records.

6.2 Acceptance Test Suites

Suite	Coverage	Test References
Authentication	Registration, login, invalid credentials, protected access	TC-AUTH-01, TC-AUTH-02
Event Management	Create, edit, publish, cancel, view, search/filter	TC-EVENT-01 to TC-EVENT-06
Registration	Capacity, deadline, duplicate, cancellation, status	TC-REG-01 to TC-REG-06
Authorization	Organizer ownership and administrator boundaries	TC-AUTHZ-01, TC-AUTHZ-02
Security	Password handling, validation, safe errors, audit	TC-SEC-01 to TC-SEC-03, TC-AUDIT-01
Performance	Listing, search, dashboard and registration timing	TC-PERF-01 to TC-PERF-03
Usability	Representative browser viewport workflows	TC-UX-01
Data Integrity	Duplicate/over-capacity edge conditions	TC-DATA-01

6.3 Acceptance Exit Criteria

- All high-priority functional requirements are implemented and verified.
- No critical security or data-integrity failure remains unresolved.
- Required acceptance test suites have been executed.
- The RTM shows traceability from requirements to design/module and test cases.
- Demonstration workflows for Participant, Organizer and Administrator complete successfully.

7. UML Use-Case Diagrams

7.1 Participant Use-Case Diagram

This diagram models the Participant's interaction with the system for account access, event discovery and registration management.

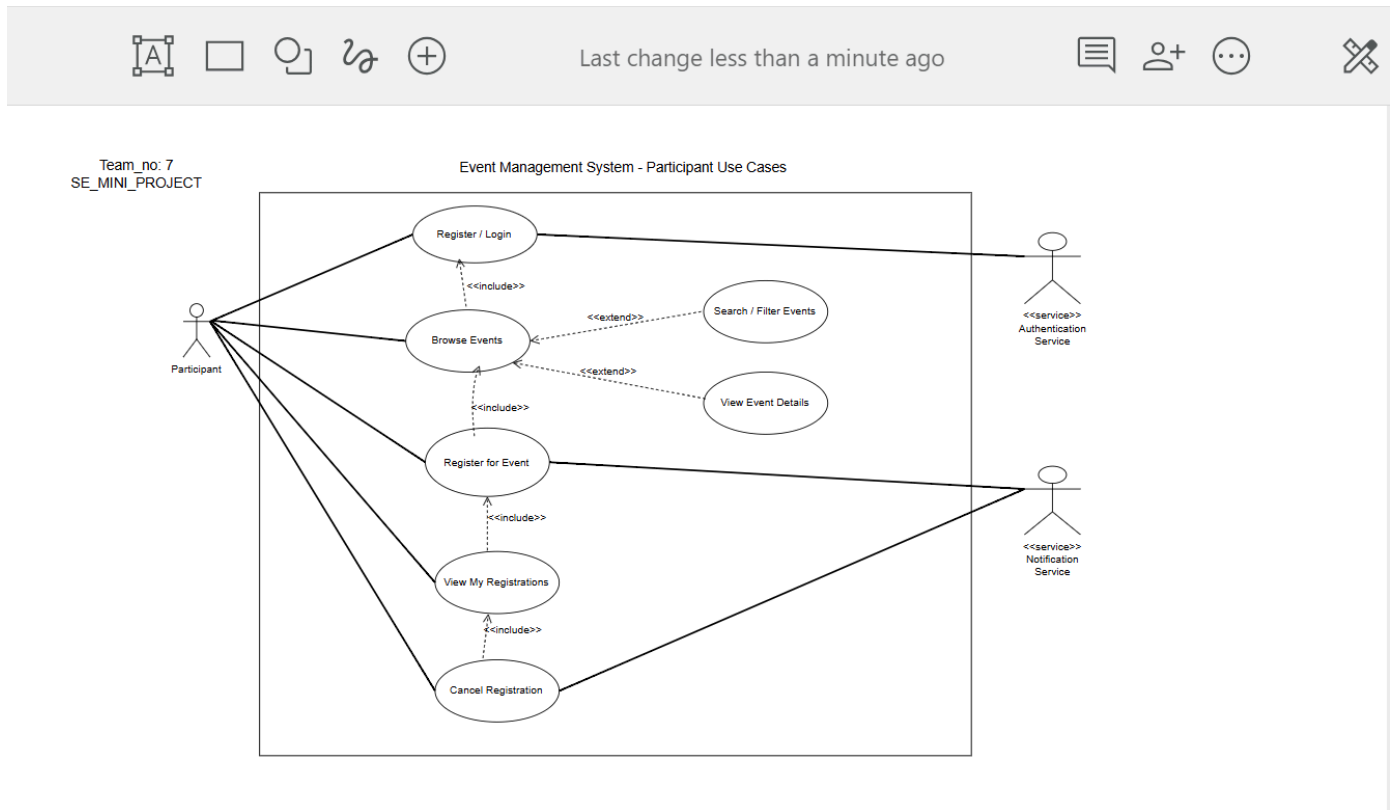


Figure 1: Participant Use-Case Diagram

7.2 Organizer & Administrator Use-Case Diagram

This diagram models Organizer event-management operations and Administrator system-management operations.

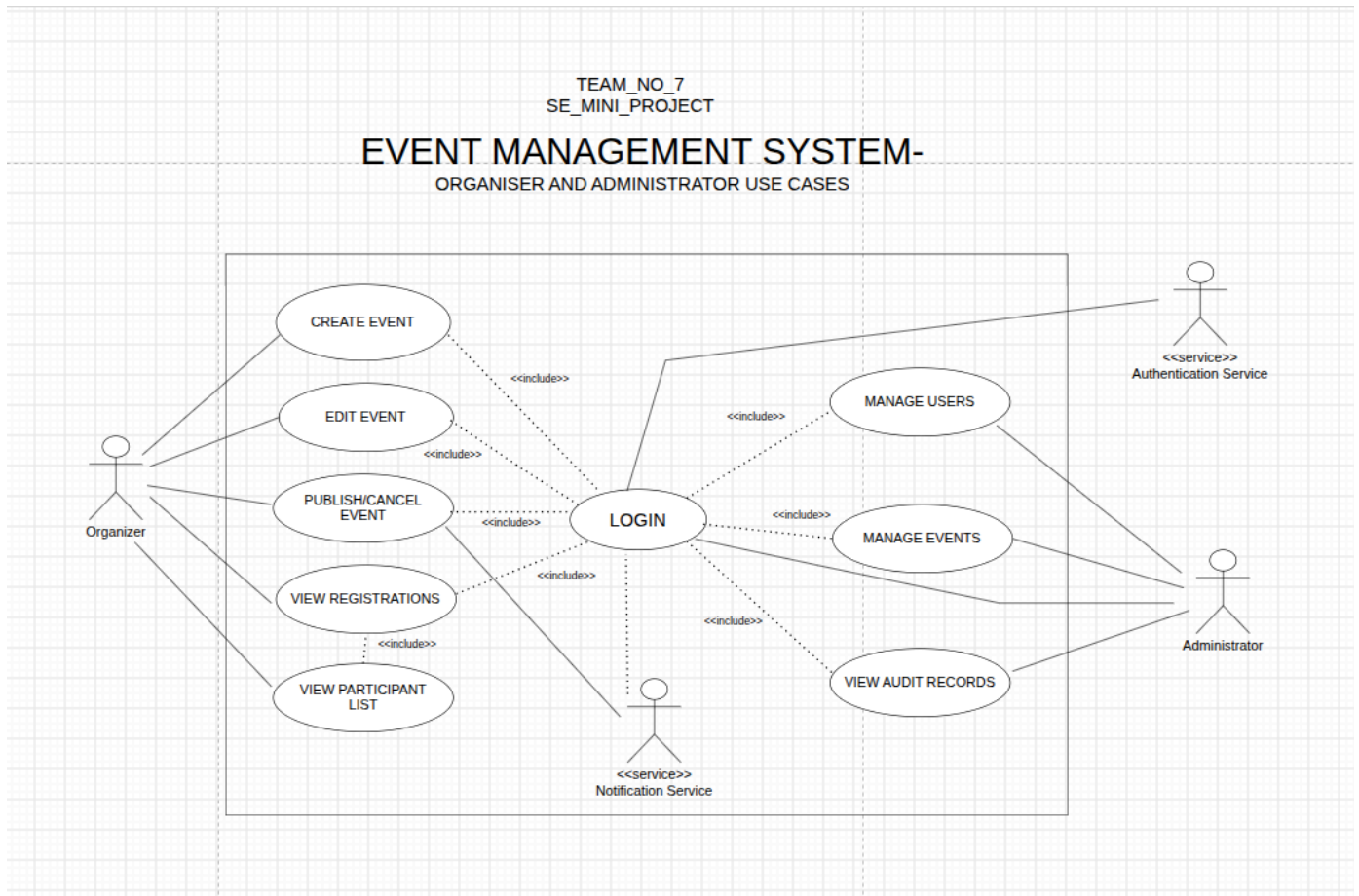


Figure 2: Organizer and Administrator Use-Case Diagram

8. Requirements Traceability Matrix (RTM)

Status values: N = Not implemented/not verified, P = Partially implemented, A = Accepted. The matrix shall be updated during development and testing.

Req ID	Requirement Short Description	Section / Design Spec	Module	Test Case(s)	Status
EMS-F-001	Create account	4.1 / DS-Auth-01	Authentication	TC-AUTH-01	N
EMS-F-002	Authenticate user	4.1 / DS-Auth-02	Authentication	TC-AUTH-02	N
EMS-F-005	Create event	4.2 / DS-Event-01	Event Module	TC-EVENT-01	N
EMS-F-007	Publish event	4.2 / DS-Event-02	Event Module	TC-EVENT-03	N
EMS-F-013	Register for event	4.4 / DS-Reg-01	Registration Module	TC-REG-01	N
EMS-F-014	Prevent duplicate registration	4.4 / DS-Reg-02	Registration Module	TC-REG-02	N
EMS-F-015	Enforce deadline/capacity	4.4 / DS-Reg-03	Registration Module	TC-REG-03	N
EMS-F-018	View registration count	4.5 / DS-Org-01	Organizer Module	TC-ORG-01	N
EMS-F-021	Manage users	4.6 / DS-Admin-01	Admin Module	TC-ADMIN-01	N
EMS-F-026	Audit important actions	4.7 / DS-Audit-01	Audit Module	TC-AUDIT-01	N
EMS-NF-001	Response-time target	5.1 / DS-Perf-01	API/Web UI	TC-PERF-01	N
EMS-NF-004	Registration data integrity	5.1 / DS-Data-01	Registration + DB	TC-DATA-01	N
EMS-SR-001	Hashed passwords	5.3.2 / DS-Sec-01	Authentication	TC-SEC-01	N
EMS-SR-002	Server-side authorization	5.3.2 / DS-Sec-02	RBAC Middleware	TC-AUTHZ-01	N
EMS-SR-005	Security audit records	5.3.2 / DS-Sec-03	Audit Module	TC-AUDIT-01	N

Appendix A: Glossary

Term	Definition
React.js	Frontend JavaScript library used for the user interface.
Node.js	JavaScript runtime used for backend application services.
Express.js	Backend web framework used to implement REST APIs.
MongoDB	Database used for persistent application data in Version 1.0.
REST API	HTTP-based interface through which frontend and backend exchange application data.
RBAC	Role-Based Access Control; permissions are assigned according to user role.
Audit Record	Timestamped record of an important administrative or security-sensitive action.

Appendix B: Division of Work & Product Ownership

The team should assign clear end-to-end ownership of major functional areas before implementation. Common UI integration, database integration, code review, testing support and final integration remain shared responsibilities.

Team Member	Functional Feature(s) Owned	Qualitative Property / Shared Responsibility
ARYA V D	Organizer Registration Management; Participant Lists; Admin User/Event Management; Notifications; Audit Records	Authorization & Auditability — role-boundary testing, audit logging, security testing, UI integration
PRANEETH P SHETTY	Register for Event; Cancel Registration; Registration History; Capacity Handling; Deadline Validation; Duplicate Registration Prevention	Reliability & Data Integrity — registration consistency, database testing, edge-case testing, API integration
SINDHU GV	User Registration; Login/Authentication; Participant Profiles; Role-Based Authorization	Security & Usability — credential handling, authentication testing, access-control testing, UI integration
PREETHI T	Create Event; Edit Event; Publish/Cancel Event; Browse Events; Search; Filter; Event Details	Performance & Maintainability — API efficiency, modular implementation, UI testing, integration

Appendix C: Other Requirements

- Application data including users, events, registrations and audit records shall be stored in MongoDB.
- The project shall use a shared repository for version control and a task board for development tracking.
- No internationalization is planned for Version 1.0.
- External payment processing is not included in Version 1.0.
- External email/SMS infrastructure is not required in Version 1.0; in-app notifications are sufficient.
- Any new major feature shall be introduced through a documented requirements change so that scope and traceability remain controlled.